

Lucy Farnik

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Profile

I'm a machine learning PhD student. Last summer I was a research intern at Jane Street. I've published papers at ICLR and ICML, completed MATS under Neel Nanda (Google DeepMind), ARENA, and various other accelerators. My previous background is in SWE – I started coding at age 7 and became a senior developer at age 18.

Research Experience

Machine Learning Research Intern <i>Jane Street</i>	June 2026 – September 2026 <i>London, UK</i>
• Developed novel architectures for learning features about the short-term market structure of US equities which help with downstream prediction tasks • Developed methods of reducing overfitting of neural networks in low signal-to-noise regimes • Explored the design space of proprietary trading systems for executing market signals	
MATS Scholar (Neel Nanda stream) <i>ML Alignment & Theory Scholars</i>	November 2023 – July 2024 <i>Berkeley, CA</i>
• Explored SAE-based circuit analysis and static analysis under Neel Nanda (Google DeepMind)	
Research Lead & Cofounder <i>Bristol AI Safety Centre (BASC)</i>	August 2023 – January 2024 <i>Bristol, UK</i>
Forecasting paradigm shifts in AI <i>Epoch AI and the Forecasting Research Institute (FRI)</i>	July 2023 – September 2023 <i>Remote</i>
Research Intern <i>University of Bristol</i>	July 2023 – September 2023 <i>Bristol, UK</i>
Researcher <i>Alignment Research Engineer Accelerator (ARENA)</i>	May 2023 – June 2023 <i>London, UK</i>
Researcher <i>AI Safety Camp</i>	February 2023 – June 2023 <i>Remote</i>

Engineering Experience

Senior Full Stack Developer <i>Longitude 103</i>	August 2019 – June 2023 <i>Remote</i>
• Became a senior developer at age 18 • Adapted to leading multiple B2B platforms in parallel, including building several products on my own, working across front end, back end, DevOps, iOS, Android, and UX • The primary project I led was bringing in ~\$300,000 in annual revenue – nearly 10x more than when I joined the company, largely due to the new features and UX improvements I built	

Publications

Jacobian Sparse Autoencoders: Sparsify Computations, Not Just Activations

L. Farnik, T. Lawson, C. Houghton, L. Aitchison

ICML 2025

- Created a new method for discovering small interpretable units of computation in neural nets

Sparse Autoencoders Can Interpret Randomly Initialized Transformers

T. Heap, T. Lawson, L. Farnik, L. Aitchison

ICLR 2026

- Demonstrated that standard methodology around SAEs can be misleading

Residual Stream Analysis with Multi-Layer SAEs

T. Lawson, L. Farnik, C. Houghton, L. Aitchison

ICLR 2025

- Studied information flow in LLMs by training a single SAE on all layers

STARC: A General Framework For Quantifying Differences Between Reward Functions

J. Skalse, L. Farnik, S. Motwani, E. Jenner, A. Gleave, A. Abate

ICLR 2024

- Created a set of reward distance metrics which exceeded the previous state of the art

Inducing Human-like Biases in Moral Reasoning Language Models

A Meek, A Karpov, S. Cho, R. Koopmanschap, L. Farnik, B. Cirstea

NeurIPS UniReps 2024

- Fine-tuned LLMs on fMRI data in order to induce more human-like behavior

Grants

Long-Term Future Fund

April 2024

- \$41,000 grant for my research on sparse circuit discovery during MATS

Lightspeed Grants

August 2023

- \$60,000 award to support my research on AI robustness

Lightspeed Grants

August 2023

- \$12,000 grant for BASC, a small research organization I co-founded

Education

University of Bristol

2023 – 2027

PhD Interactive Artificial Intelligence

Remotely from London, UK

- My PhD is focused on deep language modeling; I'm supervised by Dr. Laurence Aitchison
- 2023–24 was a qualification/coursework year which I completed with first-class honours

University of Bristol

2020 – 2023

BEng Computer Science with Innovation

Bristol, UK

- First-class honours
- Relevant units and marks: Data-Driven Computer Science (97), Artificial Intelligence (92), Machine Learning (85), Computer Architecture (96), Image Processing and Computer Vision (86)

Technical Skills

PyTorch, vLLM, Transformers, Ray, HuggingFace, WandB, Accelerate, TRL, Peft, Numpy, Redis, SLURM, HPC, Git, Python, JavaScript, Go, React, SQL, NoSQL, AWS